MPG No. 9.3.31 $4'$ [Type III, tetragonal] [T tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} T_{xx} & T_{xy} & 0 \\ T_{xy} & -T_{xx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xx} = \mathbb{T}_{v}^{(1)}$$
$$T_{xy} = \mathbb{T}_{xy}^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & -T_{xxz} \\ 0 & 0 & 0 \\ T_{yzx} & -T_{xxz} & 0 \\ T_{xxz} & T_{yzx} & 0 \\ 0 & 0 & T_{yzx} \end{bmatrix}$$

$$T_{xxz} = \mathbb{T}_{z}^{\beta(1)}$$
$$T_{yzx} = \mathbb{T}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & -T_{xxz} \\ 0 & 0 & 0 \\ T_{yzx} & \frac{T_{xxz}}{2} & 0 \\ -\frac{T_{xxz}}{2} & T_{yzx} & 0 \\ 0 & 0 & -2T_{yzx} \end{bmatrix}$$

$$T_{xxz} = 2\mathbb{M}_{xy}^{(1)}$$
$$T_{yzx} = \mathbb{M}_{v}^{(1)}$$
$$\frac{T_{xxz}}{2} = \mathbb{M}_{xy}^{(1)}$$
$$2T_{yzx} = 2\mathbb{M}_{v}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} T_{yzx} & T_{yzy} & 0 \\ T_{yzy} & -T_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzx} = \mathbb{M}_{v}^{(2)}$$
$$T_{yzy} = \mathbb{M}_{xy}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} T_{xxxx} & 0 & T_{xxzz} & 0 & 0 & T_{xxxy} \\ 0 & -T_{xxxx} & -T_{xxzz} & 0 & 0 & T_{xxxy} \\ T_{xxzz} & -T_{xxzz} & 0 & 0 & 0 & T_{zzxy} \\ 0 & 0 & 0 & T_{yzyz} & T_{yzzx} & 0 \\ 0 & 0 & 0 & T_{yzzx} & -T_{yzyz} & 0 \\ T_{xxxy} & T_{xxxy} & T_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxx} = -\mathbb{T}^{(1)}_{4v} + 2\mathbb{T}^{(1)}_{v} + 4\mathbb{T}^{(2)}_{v}$$
$$T_{xxzz} = \mathbb{T}^{(1)}_{4v} + \mathbb{T}^{(1)}_{v}$$
$$T_{xxxy} = -\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy}$$
$$T_{zzxy} = 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy}$$
$$T_{yzyz} = -\mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(2)}_{v}$$
$$T_{yzzx} = 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(2)}_{xy}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & T_{xxyy} & T_{xxzz} & 0 & 0 & T_{xxxy} \\ -T_{xxyy} & 0 & -T_{xxzz} & 0 & 0 & T_{xxxy} \\ -T_{xxzz} & T_{xxzz} & 0 & 0 & 0 & T_{zzxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -T_{xxxy} & -T_{xxxy} & -T_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxyy} = 4\mathbb{M}^{(1)}_{xyz} - 2\mathbb{T}^{(3)}_{v}$$
$$T_{xxzz} = -4\mathbb{M}^{(1)}_{xyz} - \mathbb{T}^{(3)}_{v}$$
$$T_{xxxy} = 2\mathbb{M}^{\beta(1)}_{z} + \mathbb{T}^{(3)}_{xy}$$
$$T_{zzxy} = -4\mathbb{M}^{\beta(1)}_{z} + \mathbb{T}^{(3)}_{xy}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} T_{yzyz} & T_{yzzx} & 0 \\ T_{yzzx} & -T_{yzyz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzyz} = 2\mathbb{T}^{(4)}_{v}$$
$$T_{yzzx} = 2\mathbb{T}^{(4)}_{xy}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxxy} \\ 0 & 0 & -T_{xxxy} \\ 0 & 0 & 0 \\ T_{yzyz} & T_{yzzx} & 0 \\ -T_{yzzx} & T_{yzyz} & 0 \\ 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{xxxy} = \mathbb{M}^{\beta(2)}_{z} - 2\mathbb{T}^{(5)}_{xy}$$
$$T_{yzyz} = \mathbb{M}^{(2)}_{xyz} - \mathbb{T}^{(5)}_{v}$$
$$T_{yzzx} = -\mathbb{M}^{\beta(2)}_{z} - \mathbb{T}^{(5)}_{xy}$$
$$T_{xyxy} = \mathbb{M}^{(2)}_{xyz} + 2\mathbb{T}^{(5)}_{v}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & T_{yzzx} & 0 \\ 0 & 0 & 0 & -T_{yzzx} & T_{yzyz} & 0 \\ T_{xyxx} & -T_{xyxx} & 0 & 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{M}^{(3)}_{xyz} - \mathbb{T}^{(6)}_{v}$$
$$T_{yzzx} = \mathbb{M}^{\beta(3)}_{z} + \mathbb{T}^{(6)}_{xy}$$
$$T_{xyxx} = \mathbb{M}^{\beta(3)}_{z} - 2\mathbb{T}^{(6)}_{xy}$$
$$T_{xyxy} = \mathbb{M}^{(3)}_{xyz} + 2\mathbb{T}^{(6)}_{v}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & 0 & 0 & T_{xxxy} & 0 & 0 & T_{xxyx} \\ -T_{xxyy} & -T_{xxxx} & -T_{xxzz} & 0 & 0 & T_{xxyx} & 0 & 0 & T_{xxxy} \\ T_{zzxx} & -T_{zzxx} & 0 & 0 & 0 & T_{zzxy} & 0 & 0 & T_{zzxy} \\ 0 & 0 & 0 & T_{yzyz} & T_{yzzx} & 0 & T_{yzzy} & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{yzxz} & -T_{yzzy} & 0 & T_{yzzx} & -T_{yzyz} & 0 \\ T_{xyxx} & T_{xyxx} & T_{xyzz} & 0 & 0 & T_{xyxy} & 0 & 0 & -T_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
T_{xxxx} &= -\mathbb{T}^{(1)}_{4v} + 2\mathbb{T}^{(1)}_{v} + 4\mathbb{T}^{(2)}_{v} \\
T_{xxyy} &= 4\mathbb{M}^{(1)}_{xyz} - 2\mathbb{T}^{(3)}_{v} \\
T_{xxzz} &= -4\mathbb{M}^{(1)}_{xyz} + \mathbb{T}^{(1)}_{4v} + \mathbb{T}^{(1)}_{v} - \mathbb{T}^{(3)}_{v} \\
T_{xxxy} &= 2\mathbb{M}^{\beta(1)}_{z} + \mathbb{M}^{\beta(2)}_{z} - \mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy} + \mathbb{T}^{(3)}_{xy} - 2\mathbb{T}^{(5)}_{xy} \\
T_{xxyx} &= 2\mathbb{M}^{\beta(1)}_{z} - \mathbb{M}^{\beta(2)}_{z} - \mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy} + \mathbb{T}^{(3)}_{xy} + 2\mathbb{T}^{(5)}_{xy} \\
T_{zzxx} &= 4\mathbb{M}^{(1)}_{xyz} + \mathbb{T}^{(1)}_{4v} + \mathbb{T}^{(1)}_{v} + \mathbb{T}^{(3)}_{v} \\
T_{zzxy} &= -4\mathbb{M}^{\beta(1)}_{z} + 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + \mathbb{T}^{(3)}_{xy} \\
T_{yzyz} &= \mathbb{M}^{(2)}_{xyz} - \mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(2)}_{v} - \mathbb{T}^{(5)}_{v} \\
T_{yzzx} &= -\mathbb{M}^{\beta(2)}_{z} + 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(2)}_{xy} - \mathbb{T}^{(5)}_{xy} \\
T_{yzzy} &= -\mathbb{M}^{(2)}_{xyz} - \mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(2)}_{v} + \mathbb{T}^{(5)}_{v} \\
T_{yzxz} &= \mathbb{M}^{\beta(2)}_{z} + 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(2)}_{xy} + \mathbb{T}^{(5)}_{xy} \\
T_{xyxx} &= -2\mathbb{M}^{\beta(1)}_{z} - \mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy} - \mathbb{T}^{(3)}_{xy} \\
T_{xyzz} &= 4\mathbb{M}^{\beta(1)}_{z} + 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} - \mathbb{T}^{(3)}_{xy} \\
T_{xyxy} &= \mathbb{M}^{(2)}_{xyz} + 2\mathbb{T}^{(5)}_{v}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & T_{yzzx} & 0 & T_{yzzy} & T_{yzxz} & 0 \\ 0 & 0 & 0 & -T_{yzxz} & T_{yzzy} & 0 & -T_{yzzx} & T_{yzyz} & 0 \\ T_{xyxx} & -T_{xyxx} & 0 & 0 & 0 & T_{xyxy} & 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
T_{yzyz} &= \mathbb{M}^{(3)}_{xyz} + 2\mathbb{T}^{(4)}_{v} - \mathbb{T}^{(6)}_{v} \\
T_{yzzx} &= \mathbb{M}^{\beta(3)}_{z} + 2\mathbb{T}^{(4)}_{xy} + \mathbb{T}^{(6)}_{xy} \\
T_{yzzy} &= \mathbb{M}^{(3)}_{xyz} - 2\mathbb{T}^{(4)}_{v} - \mathbb{T}^{(6)}_{v} \\
T_{yzxz} &= \mathbb{M}^{\beta(3)}_{z} - 2\mathbb{T}^{(4)}_{xy} + \mathbb{T}^{(6)}_{xy} \\
T_{xyxx} &= \mathbb{M}^{\beta(3)}_{z} - 2\mathbb{T}^{(6)}_{xy} \\
T_{xyxy} &= \mathbb{M}^{(3)}_{xyz} + 2\mathbb{T}^{(6)}_{v}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} T_{xxxx} & T_{xxxy} & 0 \\ T_{xxxy} & -T_{xxxx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & T_{yyzz} \\ T_{zzxx} & T_{zzxy} & 0 \\ T_{xxyx} & -T_{yyzz} - T_{zzxx} & 0 \\ T_{yzzx} & T_{yzzy} & 0 \\ 0 & 0 & T_{zxxz} \\ -T_{yzzy} - T_{zxxz} & T_{xyyy} & 0 \\ 0 & 0 & -\frac{T_{xxxy}}{2} - \frac{T_{xxyx}}{2} + T_{xyyy} + T_{zzxy} \end{bmatrix}$$

$$T_{xxxx} = -\mathbb{T}^{(1)}_{4v} + 2\mathbb{T}^{(1)}_{v} + 4\mathbb{T}^{(2)}_{v}$$
$$T_{xxxy} = -\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy} + \mathbb{T}^{(3)}_{xy} - 2\mathbb{T}^{(5)}_{xy}$$
$$T_{yyzz} = -\mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(1)}_{v} + \mathbb{T}^{(3)}_{v}$$
$$T_{zzxx} = \mathbb{T}^{(1)}_{4v} + \mathbb{T}^{(1)}_{v} + \mathbb{T}^{(3)}_{v}$$
$$T_{zzxy} = 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + \mathbb{T}^{(3)}_{xy}$$
$$T_{xxyx} = -\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy} + \mathbb{T}^{(3)}_{xy} + 2\mathbb{T}^{(5)}_{xy}$$
$$T_{yzzx} = 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(2)}_{xy} - \mathbb{T}^{(5)}_{xy}$$
$$T_{yzzy} = -\mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(2)}_{v} + \mathbb{T}^{(5)}_{v}$$
$$T_{zxxz} = \mathbb{T}^{(1)}_{4v} + \mathbb{T}^{(2)}_{v} + \mathbb{T}^{(5)}_{v}$$
$$T_{xyyy} = -\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy} - \mathbb{T}^{(3)}_{xy}$$